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SAVEETHA

INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Name of the Student : S.SHAMAH ALI FATHIMA

Programme: **B.E / B.Tech**

Slot : D

Course Code: **CSA1519**

Faculty : Dr.Arul Raj A.M

Course Name: **Cloud Computing and Big Data Analytics**

Date: 3/11/26

Duration: **1:30 Hr**

Max Marks: **50 Marks**

CONTINUOUS INTERNAL ASSESSMENT 1

Course Outcomes

On successful completion of the course, students will be able to:

CO	Course Outcome Statement
CO1	Apply cloud computing technologies including hardware–software evolution, virtualization, web services, and service models to develop cloud-based solutions.
CO2	Analyze virtualization architectures to identify performance and security issues in cloud computing environments.

Overall Marks Distribution :

Question No	Question	CO	Bloom's Level	Marks	
1	Q1 – Cloud Evolution and Service Models	CO1	BL3	10	
2	Q2 – Server Consolidation Energy Saving	CO1	BL3	10	
3	Q3 – VM Over-Commitme Security	CO2	BL4	10	
4	Q4 – Data Centre, SDDC, and Privacy	CO2	BL4	10	
5	Q5 – Edge Filtering and Serverless Computing	CO2	CO2	10	
Total				50	

Shamah Ali		
Student Signature	**Faculty Signature**	**Program Director**

CIA 1 TEST

Answer all the Questions

1. A media company, StreamWave, plans to modernize its IT environment. It wants (i) to move an existing content-management application to virtual machines while retaining control over the operating system and installed software, (ii) to develop and deploy a new customer-facing application without maintaining servers or runtime environments, and (iii) to subscribe to a ready-made customer relationship management (CRM) application through a browser. Explain how developments in computing hardware and Internet software have enabled this transition. Map the three requirements to IaaS, PaaS and SaaS, respectively, and explain how the broader XaaS model can support additional requirements. Justify your choices. **(10) (CO1) (BL3)**

2. A logistics company operates 15 physical servers. Each server has an average CPU utilisation of 25% and consumes 350 W. The workloads will be consolidated onto new virtualization hosts, each consuming 450 W. The target average CPU utilisation is 65% per host. The company requires N+1 fault tolerance (one additional host beyond the calculated capacity). Electricity costs ₹9 per kWh, and the systems operate 24 × 365 hours per year. Calculate (a) the number of active hosts required and the total hosts including N+1, (b) the consolidation ratio, (c) utilisation when all hosts are active and after one host fails, and (d) the annual electricity cost saving. State your assumptions.

(10) (CO1) (BL3)

Given data

Parameter	Value
Physical servers	15
Average utilisation per server	25%
Power per old server	350 W
Power per new host	450 W
Target utilisation per host	65%
Fault tolerance	N+1 (one extra host)
Electricity tariff	₹9 per kWh
Operating hours per year	24 × 365 = 8,760 h

(10) (CO1) (BL3)

3. A virtualization host has 24 physical CPU cores and 96 GB RAM, of which 8 GB is reserved for the hypervisor. Twelve VMs are deployed, each configured with 3 vCPUs and 8 GB RAM. During normal operation, each VM uses an average of 5 GB RAM and 35% of its allocated vCPU capacity. During peak operation, eight VMs use 8 GB RAM each and the remaining four use 6 GB each; all VMs use 75% of their allocated vCPU capacity. Analyse (a) the vCPU and memory over-commitment ratios, (b) average and peak memory sufficiency, (c) average and peak CPU sufficiency, and (d) possible performance and security issues with suitable remedies.

(10) (CO2) (BL4)

4. Two independent universities operate separate cloud data centres and propose to establish a cloud federation to share research-computing and digital-library services. Analyse how a conventional cloud data-centre architecture and a Software-Defined Data Centre (SDDC) can support this arrangement. Explain how Service-Oriented Architecture (SOA) enables service sharing between the universities. Examine the federation challenges and the privacy risks

associated with user presence information and federated identity. Recommend suitable technical and administrative safeguards.

(10) (CO2) (BL4)

5. A public-transport authority operates 150 road-monitoring cameras. Each camera generates one 4 MB video clip every 3 minutes, continuously. All clips are uploaded to cloud object storage through a REST API, and one serverless function is invoked for every uploaded clip. Assume 1 GB = 1000 MB, a 30-day month, a storage price of ₹1.50 per GB-month and a serverless charge of ₹0.00015 per invocation. An edge-filtering system is proposed to upload only 25% of the clips. Calculate (a) the daily data volume and average bandwidth without filtering, (b) the 30-day storage cost without filtering, (c) the daily data volume, average bandwidth, storage cost and monthly saving after edge filtering, and (d) the monthly serverless invocation cost after filtering. Explain the role of the REST API and why object storage is suitable for video clips.

(10) (CO2) (BL3)

Given data

Parameter	Value
Cameras	150
Clip size / interval	4MB every 3 minutes
Storage price	₹1 per GB-month
Edge filtering	25% of clips uploaded
Serverless price	₹0.0002 per invocation
Month / unit	30 days; 1 GB = 1000 MB

COMPUTING.

CONTINUOUS INTERNAL ASSESSMENT-1

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1) Developments in computing Hardware:

a) High performance processors:

Modern uses provides multiple cores and improved processing power.

b) Hardware virtualization:

Technologies such as Intel VT and AMD-V enable multiple virtual machines to run on a single physical server.

c) Large-spacity memory:

Increased RAM capacity supports multiple virtual machines and applications.

d) High-speed storage:

SSDs and distributed storage systems improvere application performance.

e) High speed networking:

Fible optic networks and high speed Internet enable remote access to cloud services.

2) Developments of Internal Softwares:

Important developments include

* Web browser and web applications.
* REST APIs and web services
* Dashboarded application architectures.

3) Mapping Requirements to cloud Service Models.

	Requirement	Cloud Model	Justification.
i)	Move existing content management application to VMs while retaining OS control	Iaas.	Provide virtual machines storage and networking with OS level control
ii)	Develop and deploy a new customer facing application.	PaaS.	Provides manage runtime middleware and deployment environment

4) Infrastructure as a Service (Iaas).

Iaas provides virtualized computing resources such as.

* Virtual machine
* storage
* Networking

al firewalls.

atform as a service (Paas).

The provider manages

* Operating systems
* Runtime environments
* Middleware
* Infrastructure

The customer focused on application code.

Exc: Google App Engine.

6) software as a service (Saas).

Saas provides complete application through the Internet. The provider manages the application.

7) Broader Xaas Model.

The Broader Xaas Model provides a real time Broad classification.

* SIaas - Storage as a service
* DBaas - Database as a service.
* Faas - Function as a Service

VER CONSOLIDATION AND ENERGY SAVING.

Parameter	Value.
* Average CPU utilization	25 %.
* Power consumption for service	550 W.
* Non virtualization host power	480W
* Efficieng cost	₹9 KWh.

a) Number of Active HOSTS and Total Hosts.

Total unordered. $W = 15 \times 25\%$

$W = 3.75$.

At 65% forget utilization.

$$N = \frac{3.75}{0.65}$$

$N = 5.70$

N = 6 active hosts

including NH:

$N_{total} = 6 + 1$

$N_{total} = 7$ hosts

b) Consolidation Ratio:

$$CR = \frac{\text{Original ismov}}{\text{New hosts}}$$

$$CR = \frac{15}{6} \qquad CR = 2.5 : 1$$

Including the additional standby host

$$CR = \frac{15}{7} = 2.14 : 1$$

Active host consolidation ratio = 2.5:1

Total host consolidation ration = 2.14:1

c) CPU utilization:

Total workloaded = 3.75 servers.

when all 4 hosts are active

$$u = \frac{3.75}{4} \times 100$$

$$u = 53.57\ \%$$

when one host fails.

Six hosts

$$u = \frac{3.75}{6} \times 100$$

$$u = 62.5\ \%$$

Annual Electricity Cost Saving.

$P_{old} = 15 \times 350$

$P_{old} = 5250\ W = 5.25\ kW.$

Annual energy

$E_{old} = 5.25 \times 8760$

$E_{old} = 45599\ 0\ kwh$

Annual cost:

$cold = 45999 0 \times 9$

$Cold = 41390.$

Neo power system

$P_{new} = 7 \times 450$

$= 3.15\ kw.$

Anual Energy

$E_{new} = 3.15 \times 8760$

$= 27594\ kwh.$

Annual lost.

$C_{new} = 24834 6$

Annual savings.

$$S = 413910 - 44346$$

$$S = 1,65,564.$$

Percentage saving:

$$\frac{165564}{413910} \times 100$$

$$= 40\%.$$

3) VIRTUAL CPU and MEMORY OVER UNTERMINANT ALALYSIS.

9) CPU & memory over commitment ratios.

Total allocated $V_{CPU} = 12 \times 3 = 36.$

Physical CPU Cores $= 24$

$$R_{CPO} = \frac{36}{24}$$

$$\boxed{R_{CPO} = 65\%}$$

Average and Peak Memory sufficiency.

Total memory usage

$M. = 12 \times 5$

$M = 60 GB.$

Available RAM = 88 GB.

$88 - 60 = 28 GB.$

Normal operations is memory sufficient.

Peak operation.

Eight VMs use 8 GB each.

$8 \times 8 = 64 GB.$

four VMs use 6 GB each.

$4 \times 6 = 24 GB.$

Total $M_{peak} = 64 + 24$

$M_{peak} = 88 GB.$

c) Average and Peak CPU sufficiency.

Each VM uses 35% & 3 vCPUs

$CPU_{VM} = 3 \times 0.35$

$CPU_{VM} = 1.05$.

For 12 VMs.

$CPU_{annual} = 12 \times 1.05$

$CPU_{annual} = 12.6$

Normal CPU demand = 12.6 cores

$U = \frac{126}{24} \times 100$

$= 52.5\%$

At 75% & 3 VCPUs.

$CPU_{VM} = 3 \times 0.75$.

$CPU_{VM} = 2.25$.

For 12 VMs

$CPU_{peak} = 12 \times 2.25$

$CPU_{peak} = 27$ cores.

$U = \frac{27}{24} \times 100$

$U = 112.5\%$

..edies.

* Add more physical CPU uses.
* Release unecessary CPU allocation.
* Path and physical RAM.

4) CLOUD FEDERATION BETWEEN TWO UNIVERSITIES.

a) Conventional cloud Data center Architecture

* Physical Server
* Merge System.
* Hypervisor.

Each university operates its own infrastructure.

b) SDDC Architecture:

SDDC computes storage and networking.

Benefits:

i) Dynamic resource allocation.
ii) Automated workloaded placement.
iii) Centralized monitoring.

c) Role of SOA:

SOA enables applications to communicate through standardize interfaces.

Example.

i) University B requests research computing resources.

ii) university A exposes a research computing service.

iii) The service authentication the request-

iv) computing resource allocations.

d) Federation challenges:

Different hardware and software environments may cause a different federation.

* Resource handling
* Data Privacy.

e) Privacy Risks & User Presence Information.

Risk include.

* Tracking researchers.
* Revealing work schedules.

guards.

* Collect minimal presense information.
* obtain consent
* Restrict access.
*

e) Privacy Risks & Federated Identity.

Risks include:

* Identity impersonation.
* Unauthorized access
* Credential theft.

Technical Safeguards.

* MFA.
* Encryption
* Audit logging.

5) Edge filtering and struggles compu

a) Daily Data volume and bandwidth without fil

$20 \times 24 = 480$.

Total clips per day

$480 \times 150 = 72000$.

Daily data volume.

$D = 72000 \times 4$

$D = 288000$ MB

$\boxed{D = 285\ GB/day}$

Average bandwidth.

$$B = \frac{288 \times 100 \times 8}{86 \times 100}$$

$\boxed{B = 26.67\ Mbps}$.

b) 30-day storage cost without filtering.

Step 1: Monthly data volume.

$288 \times 30 = 8640$ GB.

Step 2: storage cost.

$C = 8640 \times 1.50$

$\boxed{C = ₹\ 12,960}$

volume, average bandwidth, storage cost & saving after edge filtering.

Step 1: Daily clips uploaded.

$$12000 \times \frac{25}{100}$$

$$= 18000 \text{ clips/day}$$

Step 2: Daily data volume.

$$18000 \times 4 \div 72000 \text{ MB}$$

$$\boxed{D = 72 \text{ GB/day}}$$

Step 3: Average bandwidth.

$$B = \frac{72 \times 1000 \times 8}{86400}$$

$$\boxed{B = 6.67 \text{ Mbps}}$$

Step 4: Monthly storage volume.

$$12 \times 30 = 2160 \text{ GB}.$$

Step 5: storage cost after filtering.

$$C = 2160 \times 1.50$$

$$\boxed{C = ₹ 3240}$$

Step 6: Monthly storage saving.

$$\text{saving} = 12960 - 3240$$

$$\boxed{\text{Saving} = ₹ 9720}$$

d) Monthly serverless invocation cost after filtering.

Step 1: Daily serverless invocations = 18000

Step 2: Monthly invocations.

$$18000 \times 30 = 540000.$$

Step 3: Monthly invocation cost.

$$C = 540000 \times 0.00015$$

$$\boxed{C = ₹81}$$

Final Ans:

Parameter	without filtering	After filtering.
* Daily data volume	288 GB	72 GB.
* Average bandwidth	26.67 Mbps	6.67 Mbps
* Monthly storage volume	8640 GB	2160 GB.
* Storage cost	₹ 12,960	₹ 3,240.